

CERTIFICATE OF COMPLIANCE - RESIDENTIAL PERFORMANCE COMPLIANCE METHOD

Project Name: John Vo Duplex RES ADU
Calculation Description: John Vo Duplex Res-ADU

Calculation Date/Time: 2024-10-16T23:32:40-07:00
Input File Name: 4357 Logan Ave.rbd22

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GENERAL INFORMATION											
01	Project Name	John Vo Duplex RES ADU									
02	Run Title	John Vo Duplex Res-ADU									
03	Project Location	4367-B Logan Ave									
04	City	San Diego, CA		05	Standards Version	2022					
06	Zip code	92113		07	Software Version	CBECC-Res 2022.3.1					
08	Climate Zone	7		09	Front Orientation (deg/ Cardinal)	270					
10	Building Type	Single family		11	Number of Dwelling Units	1					
12	Project Scope	Newly Constructed		13	Number of Bedrooms	3					
14	Addition Cond. Floor Area (ft²)	0		15	Number of Stories	2					
16	Existing Cond. Floor Area (ft²)	n/a		17	Fenestration Average U-factor	0.3					
18	Total Cond. Floor Area (ft²)	1200		19	Glazing Percentage (%)	14.50%					
20	ADU Bedroom Count	n/a		21	ADU Conditioned Floor Area	n/a					
22	Fuel Type	Natural gas		23	No Dwelling Unit:	No					

COMPLIANCE RESULTS	
01	Building Complies with Computer Performance
02	This building incorporates features that require field testing and/or verification by a certified HERS rater under the supervision of a CEC-approved HERS provider.
03	This building incorporates one or more Special Features shown below

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CA Building Energy Efficiency Standards - 2022 Residential Compliance

Registration Date/Time: 10/16/2024 23:33
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Schema Version: rev 20220901

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ENERGY DESIGN RATINGS						
	Energy Design Ratings			Compliance Margins		
	Source Energy (EDR1)	Efficiency ¹ EDR (EDR2efficiency)	Total ² EDR (EDR2total)	Source Energy (EDR1)	Efficiency ¹ EDR (EDR2efficiency)	Total ² EDR (EDR2total)
Standard Design	31.1	45.6	31.8			
Proposed Design	25.9	37.9	23.9	5.2	7.7	7.9
RESULT ³ : PASS						
¹ Efficiency EDR includes improvements like a better building envelope and more efficient equipment ² Total EDR includes efficiency and demand response measures such as photovoltaic (PV) system and batteries ³ Building complies when source energy, efficiency and total compliance margins are greater than or equal to zero and unmet load hour limits are not exceeded						
• Standard Design PV Capacity: 2.22 kWdc						

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ENERGY USE SUMMARY						
Energy Use	Standard Design Source Energy (EDR1) (kBtu/ft² - yr)	Standard Design TDV Energy (EDR2) (kTDV/ft² -yr)	Proposed Design Source Energy (EDR1) (kBtu/ft² -yr)	Proposed Design TDV Energy (EDR2) (kTDV/ft² -yr)	Compliance Margin (EDR1)	Compliance Margin (EDR2)
Space Heating	0.38	1.76	0.24	1.08	0.14	0.68
Space Cooling	0.34	8.34	0.26	6.64	0.08	1.7
IAQ Ventilation	0.42	4.46	0.42	4.46	0	0
Water Heating	1.78	19.96	1.45	16.56	0.33	3.4
Self Utilization/Flexibility Credit			0	0	0	0
Efficiency Compliance Total	2.92	34.52	2.37	28.74	0.55	5.78
Photovoltaics	-2.08	-52.67	-2.92	-62.04		
Battery			0	0		
Flexibility			0			
Indoor Lighting	0.8	8.31	0.8	8.31		
Appl. & Cooking	1.88	21.31	1.88	21.29		
Plug Loads	4.55	48	4.55	48		
Outdoor Lighting	0.2	1.86	0.2	1.86		
TOTAL COMPLIANCE	8.27	61.33	6.88	46.16		

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ENERGY USE INTENSITY				
	Standard Design (kBtu/ft² - yr)	Proposed Design (kBtu/ft² - yr)	Compliance Margin (kBtu/ft² - yr)	Margin Percentage
Gross EUI ¹	13.64	12.87	0.77	5.65
Net EUI ²	3.67	1.49	2.18	59.4
Notes 1. Gross EUI is Energy Use Total (not including PV) / Total Building Area. 2. Net EUI is Energy Use Total (including PV) / Total Building Area.				

REQUIRED PV SYSTEMS											
01	02	03	04	05	06	07	08	09	10	11	12
DC System Size (kWdc)	Exception	Module Type	Array Type	Power Electronics	CFI	Azimuth (deg)	Tilt Input	Array Angle (deg)	Tilt: (x in 12)	Inverter Eff. (%)	Annual Solar Access (%)
2	NA	Standard (14-17%)	Tracking (one axis)	none	false	180	Pitch	18.43	4	96	98

REQUIRED SPECIAL FEATURES											
The following are features that must be installed as condition for meeting the modeled energy performance for this computer analysis.											
• PV System: 2 kWdc • PV Array Type: Tracking (one axis) • Compact distribution system basic credit • Northwest Energy Efficiency Alliance (NEEA) rated heat pump water heater; specific brand/model, or equivalent, must be installed											

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HERS FEATURE SUMMARY											
The following is a summary of the features that must be field-verified by a certified HERS Rater as a condition for meeting the modeled energy performance for this computer analysis. Additional detail is provided in the building tables below. Registered CF2Rs and CF3Rs are required to be completed in the HERS Registry											
• Quality insulation installation (QII) • Indoor air quality ventilation • Kitchen range hood • Minimum Airflow • Verified Refrigerant Charge • Fan Efficacy Watts/CFM • Duct leakage testing											

BUILDING - FEATURES INFORMATION						
01	02	03	04	05	06	07
Project Name	Conditioned Floor Area (ft²)	Number of Dwelling Units	Number of Bedrooms	Number of Zones	Number of Ventilation Cooling Systems	Number of Water Heating Systems
John Vo Duplex RES ADU	1200	1	3	1	0	1

ZONE INFORMATION						
01	02	03	04	05	06	07
Zone Name	Zone Type	HVAC System Name	Zone Floor Area (ft²)	Avg. Ceiling Height	Water Heating System 1	Status
Conditioned	Conditioned	HVAC	1200	9	DHW System	New

OPAQUE SURFACES							
01	02	03	04	05	06	07	08
Name	Zone	Construction	Azimuth	Orientation	Gross Area (ft²)	Window and Door Area (ft2)	Tilt (deg)
Wall-Front	Conditioned	R15 Ext Wall	270	Front	290	64	90
Wall-Left	Conditioned	R15 Ext Wall	0	Left	527	60	90
Wall-Right	Conditioned	R15 Ext Wall	180	Right	527	56	90
Wall-Back	Conditioned	R15 Ext Wall	90	Back	290	34	90
Ceiling (below attic) 1	Conditioned	Ceiling R30	n/a	n/a	560	n/a	n/a

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ATTIC							
01	02	03	04	05	06	07	08
Name	Construction	Type	Roof Rise (x in 12)	Roof Reflectance	Roof Emittance	Radiant Barrier	Cool Roof
Attic	ROOF Asphalt	Ventilated	4	0.1	0.8	No	No

FENESTRATION / GLAZING													
01	02	03	04	05	06	07	08	09	10	11	12	13	14
Name	Type	Surface	Orientation	Azimuth	Width (ft)	Height (ft)	Mult.	Area (ft²)	U-factor	U-factor Source	SHGC	SHGC Source	Exterior Shading
01	Window	Wall-Front	Front	270	5	4	1	20	0.3	NFRC	0.23	NFRC	Bug Screen
21	Window	Wall-Front	Front	270	4	4	1	16	0.3	NFRC	0.23	NFRC	Bug Screen
22	Window	Wall-Front	Front	270	2	4	1	8	0.3	NFRC	0.23	NFRC	Bug Screen
06	Window	Wall-Left	Left	0	4	4	1	16	0.3	NFRC	0.23	NFRC	Bug Screen
07	Window	Wall-Left	Left	0	2.5	4	1	10	0.3	NFRC	0.23	NFRC	Bug Screen
08	Window	Wall-Left	Left	0	2.5	4	1	10	0.3	NFRC	0.23	NFRC	Bug Screen
26	Window	Wall-Left	Left	0	4	4	1	16	0.3	NFRC	0.23	NFRC	Bug Screen
27	Window	Wall-Left	Left	0	2	4	1	8	0.3	NFRC	0.23	NFRC	Bug Screen
02	Window	Wall-Right	Right	180	2	4	1	8	0.3	NFRC	0.23	NFRC	Bug Screen
03	Window	Wall-Right	Right	180	3.5	4	1	14	0.3	NFRC	0.23	NFRC	Bug Screen
23	Window	Wall-Right	Right	180	3.5	4	1	14	0.3	NFRC	0.23	NFRC	Bug Screen
04	Window	Wall-Back	Back	90	3	1	1	3	0.3	NFRC	0.23	NFRC	Bug Screen
05	Window	Wall-Back	Back	90	4	4	1	16	0.3	NFRC	0.23	NFRC	Bug Screen
24	Window	Wall-Back	Back	90	3	1	1	3	0.3	NFRC	0.23	NFRC	Bug Screen

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